

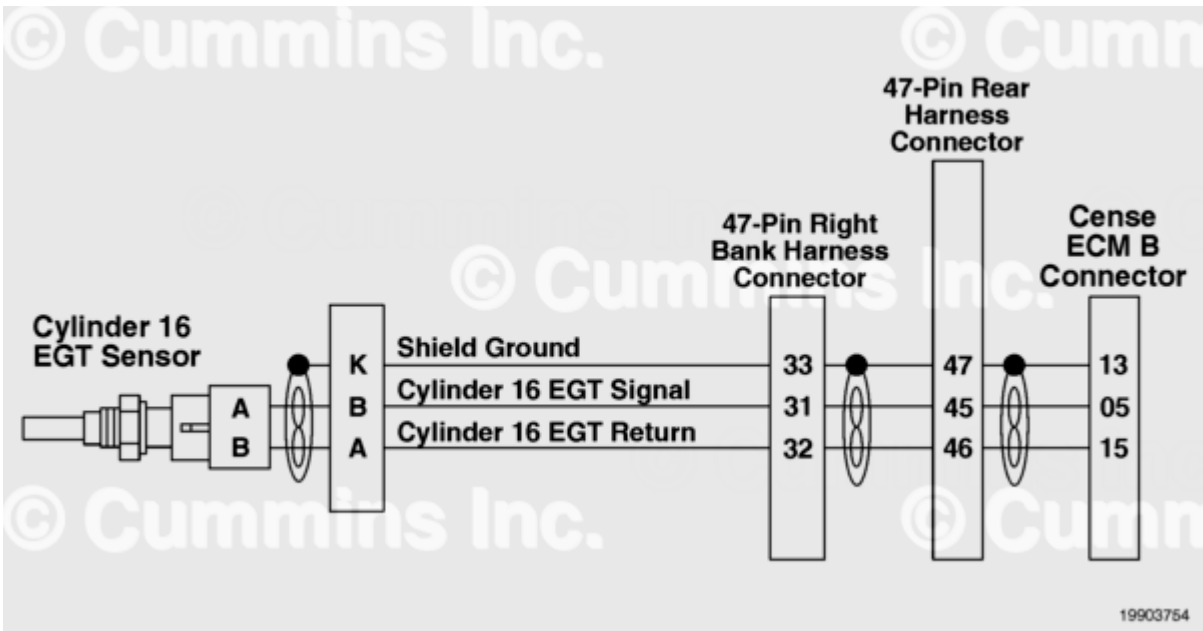
Fault Code: 638

Exhaust Gas Temperature Deviation Low for Cylinder 16 - Data Valid But Below Normal Operating Range - Least Severe Level

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Overview

Codes	Reason	Effect
Fault Code: 638 PID(P): SPN: 1338 FMI: 0/17 Lamp: Maintenance SRT:	Exhaust Gas Temperature Deviation Low for Cylinder 16 - Data Valid But Below Normal Operating Range - Least Severe Level.	Possible low power. No engine protection for exhaust gas temperature.



Exhaust Gas Temperature Sensor Circuit Cylinder 16 - QSK45 and QSK60 Engines

Circuit Description

The exhaust gas temperature sensor circuit cylinder 16 monitors exhaust gas temperature and passes information to the engine control module (ECM) through the engine harness.

Component Location

The exhaust gas temperature sensor, cylinder 16, is located in the exhaust manifold at the cylinder head 16 to exhaust manifold interface.

Shoptalk

There are multiple CENSE™ ECMs for the engine models included in this manual. The ECM model displays when INSITE™ electronic service tool is connected. When troubleshooting a fault code, use the ECM model displayed in INSITE™ electronic service tool to determine which cylinder is affected. For engines with the present CM2330 ECM, the cylinder numbering sequence is described in the General Engine procedure of Section V in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 018-015 in Section V.

(/qs3/pubsys2/xml/en/procedures/56/56-018-015-tr.html)

The exhaust gas temperature sensor circuit cylinder 16 measures the exhaust temperature of cylinder 16. The ECM monitors the temperature and compares it to the exhaust gas temperatures of other cylinders.

It is possible that the exhaust gas temperature sensor 5 volt supply can be open and a fault will **not** be set. If this happens, the temperature of the exhaust gas temperature sensor will read 474°C [885°F] for the entire bank of sensors connected to that exhaust gas temperature sensor converter. See the Engine Performance Troubleshooting Tree in Section TT of the QSK45 and QSK60 Service Manual, Bulletin 4021530, to troubleshoot this condition.

Possible causes of this fault include:

- Restricted aftercooler.
- Damaged valves, rings, or piston.
- Injector damage.

Refer to Troubleshooting Fault Code t05-638. (/qs3/pubsys2/xml/en/procedures/81/81-t05-638.html)